

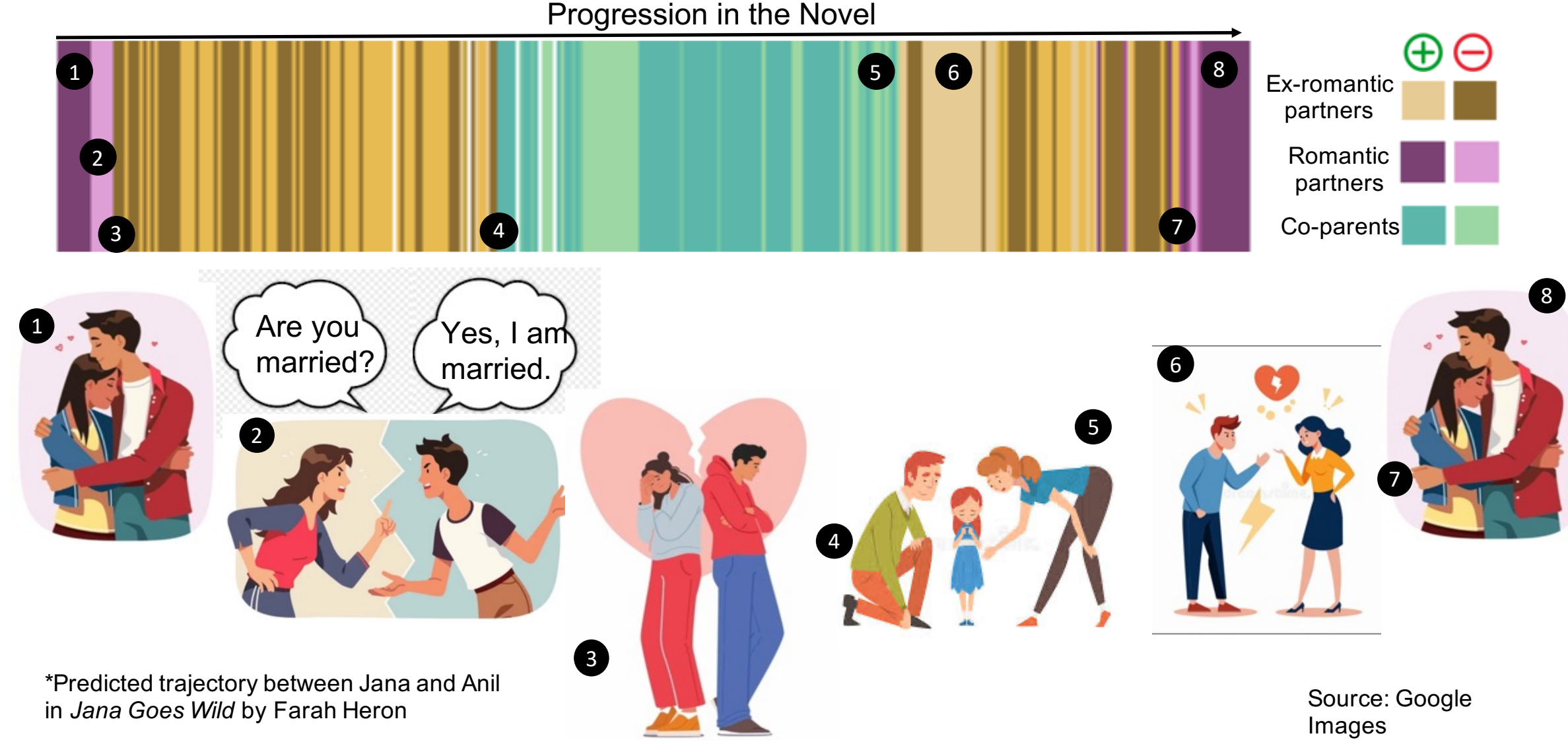
Tracking Evolving Relationship Between Characters in Books in the Era of Large Language Models

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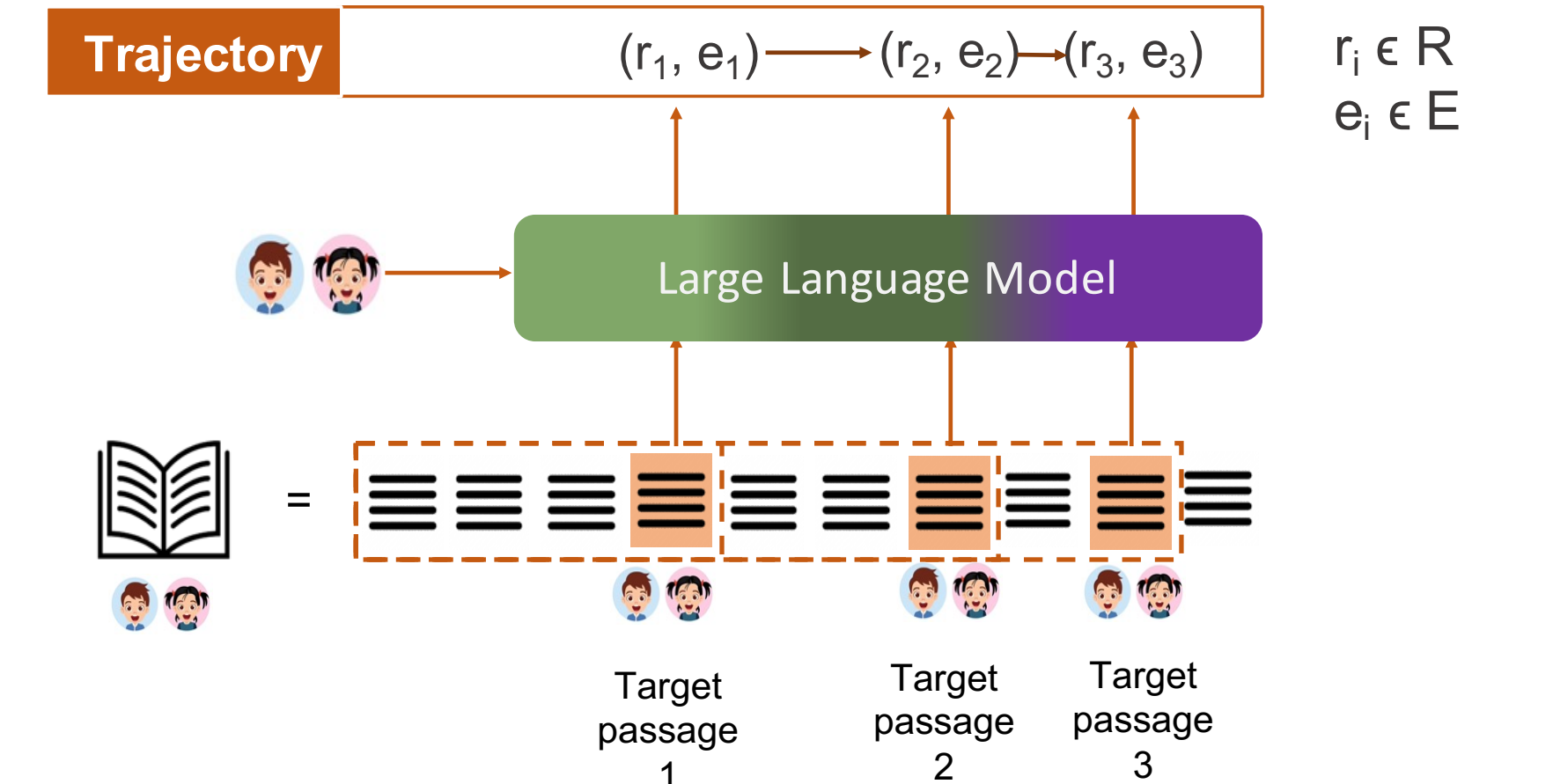
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1. Motivation



1. How can we **characterize evolution in the relationship between characters** in a book?
2. How can we **use LLM's zero-shot reasoning capabilities (without gold labels)** to track such fine-grained evolution in relationships?

2. Task of Tracking Evolution in Relationship



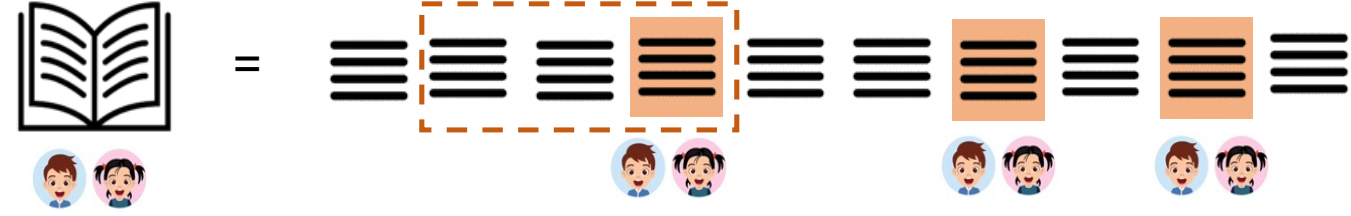
- **Relationships (R)**: Romantic, Social, Anti-Social (Jurgens et al., 2023)
- **Evolution (E)** (fine-grained; local trend): Positive, Negative, Stable (Deri et al., 2018)

3. Proposed Strategies

- Assess LLM's zero-shot social reasoning capabilities by predicting the relationship and evolution type given passages until a target passage

Use LLM with a *large context window* to provide **all passages until** that can fit in its window

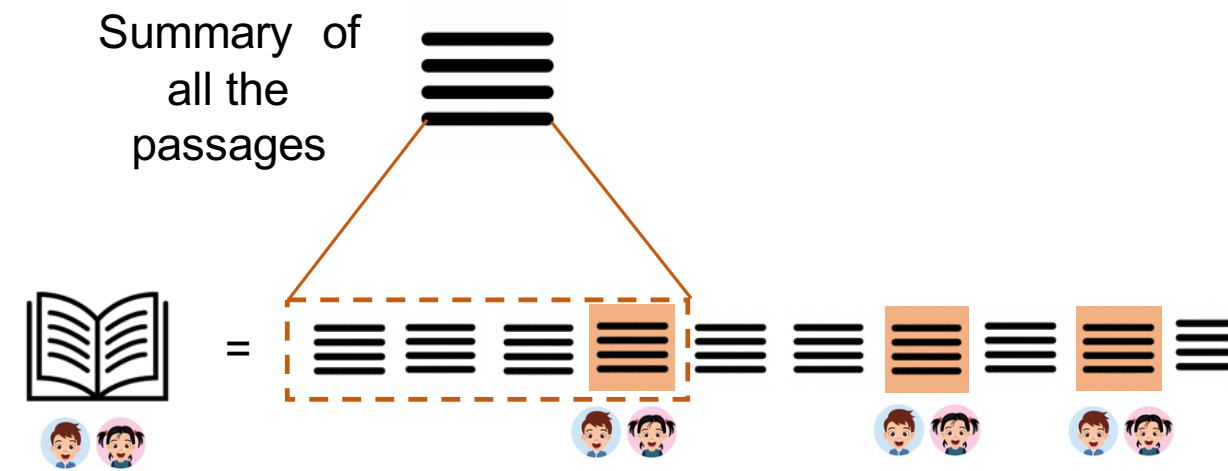
- High granularity



Complete Passages

Use LLM with a *relatively smaller context window*

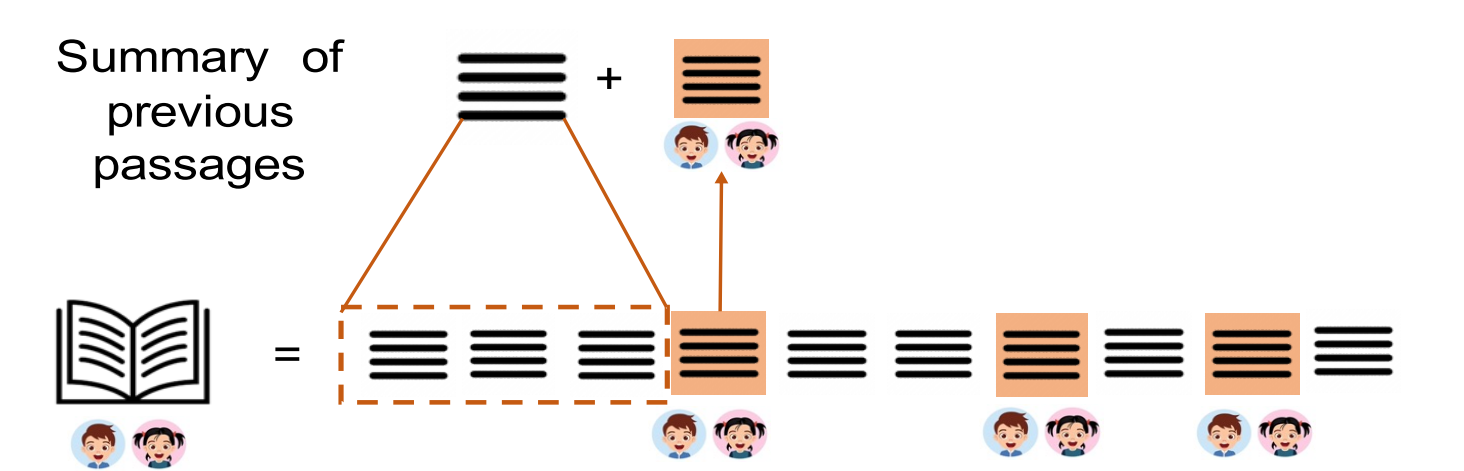
- Summary of **type and nature of evolution in relationship** between characters
- Low granularity



Complete Summary

Use LLM with a *relatively smaller context window*

- Summary of previous passages +
- Mid granularity

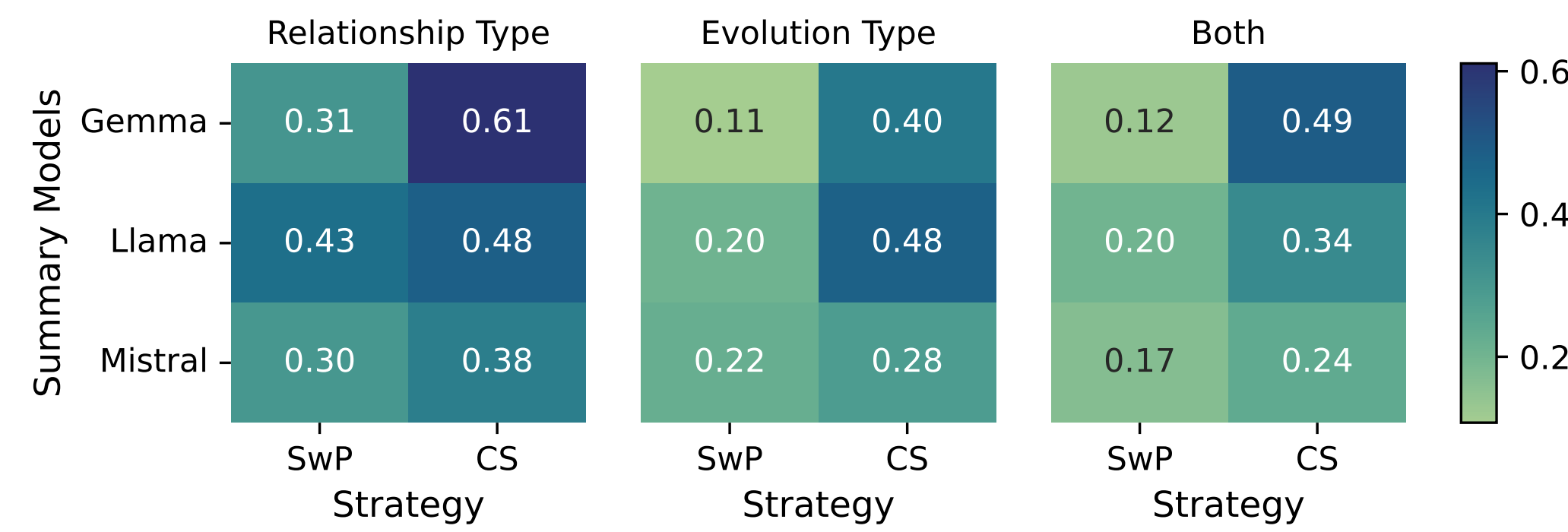


Summary with Passage

4. Experimental Setup

- Prompt Llama, Mistral, and Gemma in a zero-shot setting
- 11 romance genre books with different tropes.
- Preprocess books *i.e.*, coreference resolution + substitution and chunking
- **Evaluation (no gold labels)**
 - *Automatic*: Agreement between predictions from different LLMs per strategy
 - *Manual*: a). Global - Predict trope given predicted trajectory; and b). Local - Accuracy of consensus predictions

5. Agreement between Predictors and Strategies



Krippendorff's α between different predictions for Summary with Passage (SwP) and Complete Summary (CS) strategies using summaries generated from various summary models.

Strategy	Relationship Type	Evolution Type	Both
Summary with Passage	0.50	0.44	0.38
Complete Summary	0.47	0.12	0.17

Krippendorff's α between consensus predictions from each strategy with that from Complete Passages strategy

- Evolution type prediction is more challenging than relationship type
- $\alpha_{CS} > \alpha_{SwP}$ due to the presence of direct evidence for evolution in relationship in summaries

- Summary with Passage is a better approximation of Complete Passages strategy than Complete Summary

6. Manual Evaluation

Strategy	Relationship Type	Evolution Type	Both
Summary with Passage	86.04 (0.70)	44.18 (0.26)	39.54 (0.31)
Complete Summary	88.37 (0.79)	70.93 (0.58)	67.44 (0.61)

Local-level: Accuracy (agreement between humans) of consensus predictions from LLMs averaged across two human annotators.

- **Local:** Similar agreement between humans as observed between LLMs
- **Global:** Trope can be predicted with the highest accuracy when LLM has access to all the passages
- **Global:** Summary with passage regularizes any loss in information in the complete summary

Strategy	Summarizer	Accuracy (%)
Majority	N/A	54.54
Complete Passages	N/A	63.64
Summary with Passage	Llama	27.27
	Mistral	18.18
	Gemma	18.18
	Overall	21.21
Complete Summary	Llama	9.09
	Mistral	9.09
	Gemma	0.00
	Overall	6.06

Global-level: Accuracy of trope prediction given trajectory visualization

7. Findings from Qualitative Analysis

- Maintaining a running summary helps resolve the ambiguity between different relationship types
- Uncertainty in relationship or evolution type prediction results in disagreements between humans
- LLMs rely on surface-level cues, tend to resolve pronouns to the character in question in the absence of an explicit mention, and are sensitive to subtle changes in the context

8. Conclusion

- Agreement analysis shows that **providing a running summary of type and nature of evolution in relationship between characters along with a passage is a better approximation of a strategy that uses all passages than a complete summary in predicting evolution in relationship between characters**
- Low-to-moderate agreement between LLMs as well as between humans shows the difficulty of the task
- Qualitative analysis raises questions on LLM's true understanding of social relationships